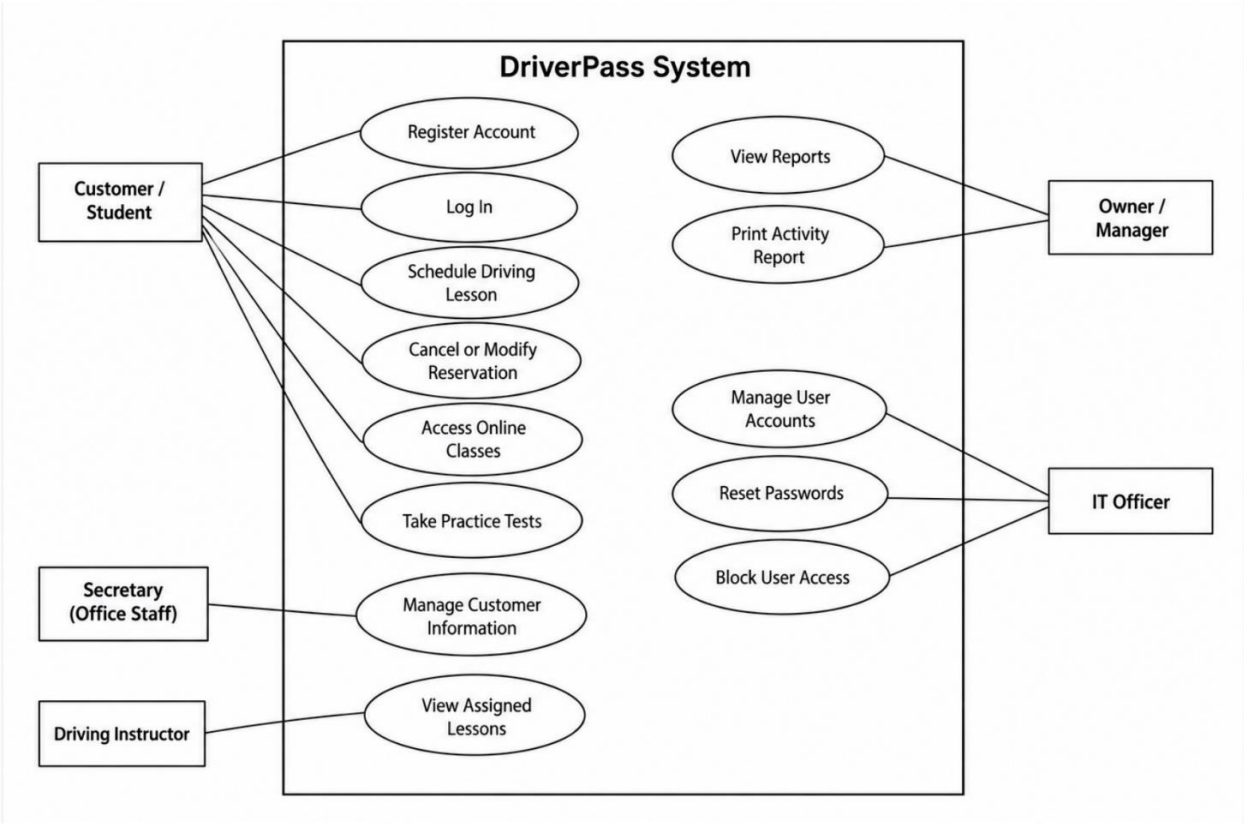
CS 255 System Design Document

Project Two — System Analysis and Design

Prepared by: Yunus Parvez Khan
Course: CS-255 — System Analysis and Design
Institution: Southern New Hampshire University
Date: October 2026

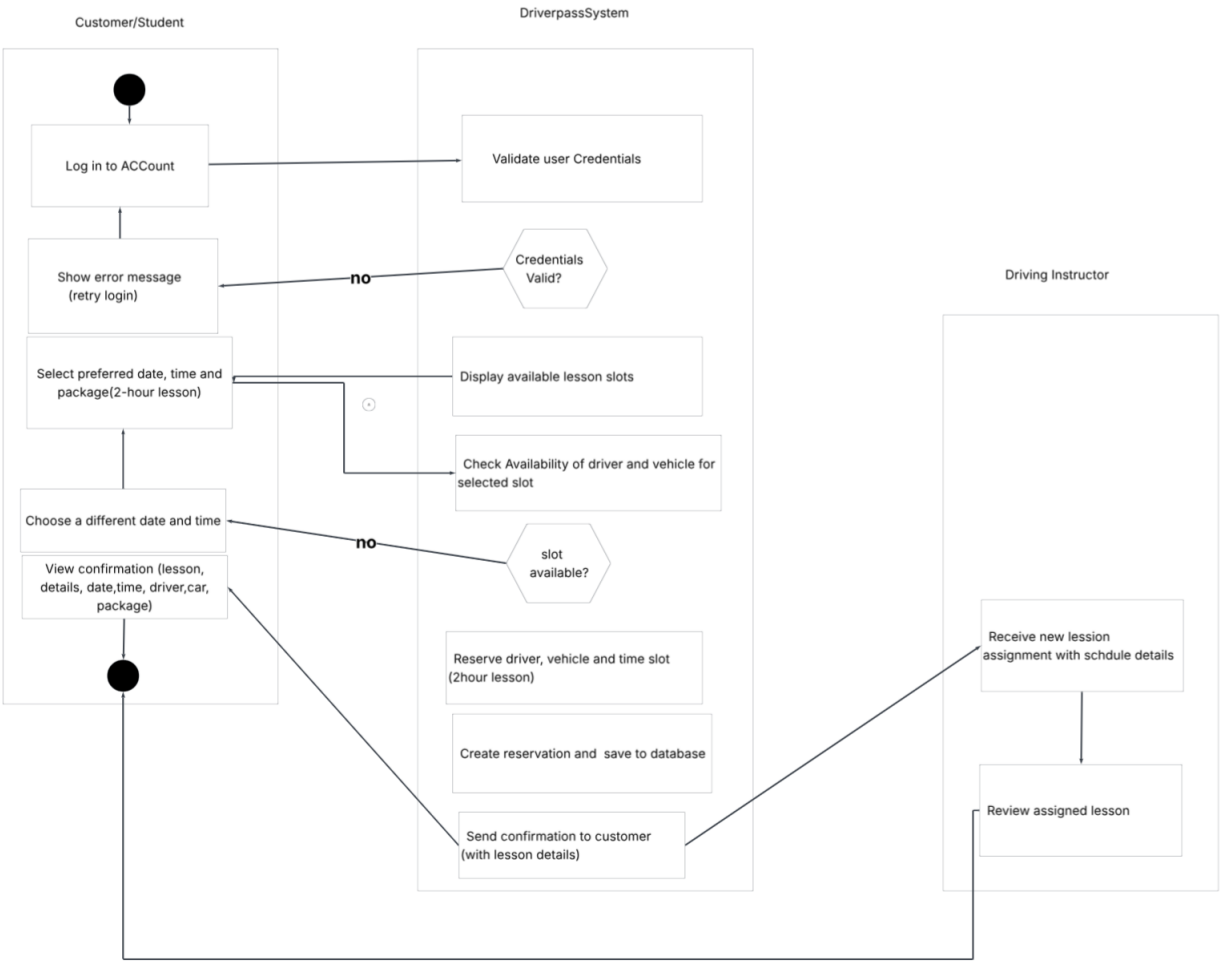
UML Diagrams

UML Use Case Diagram

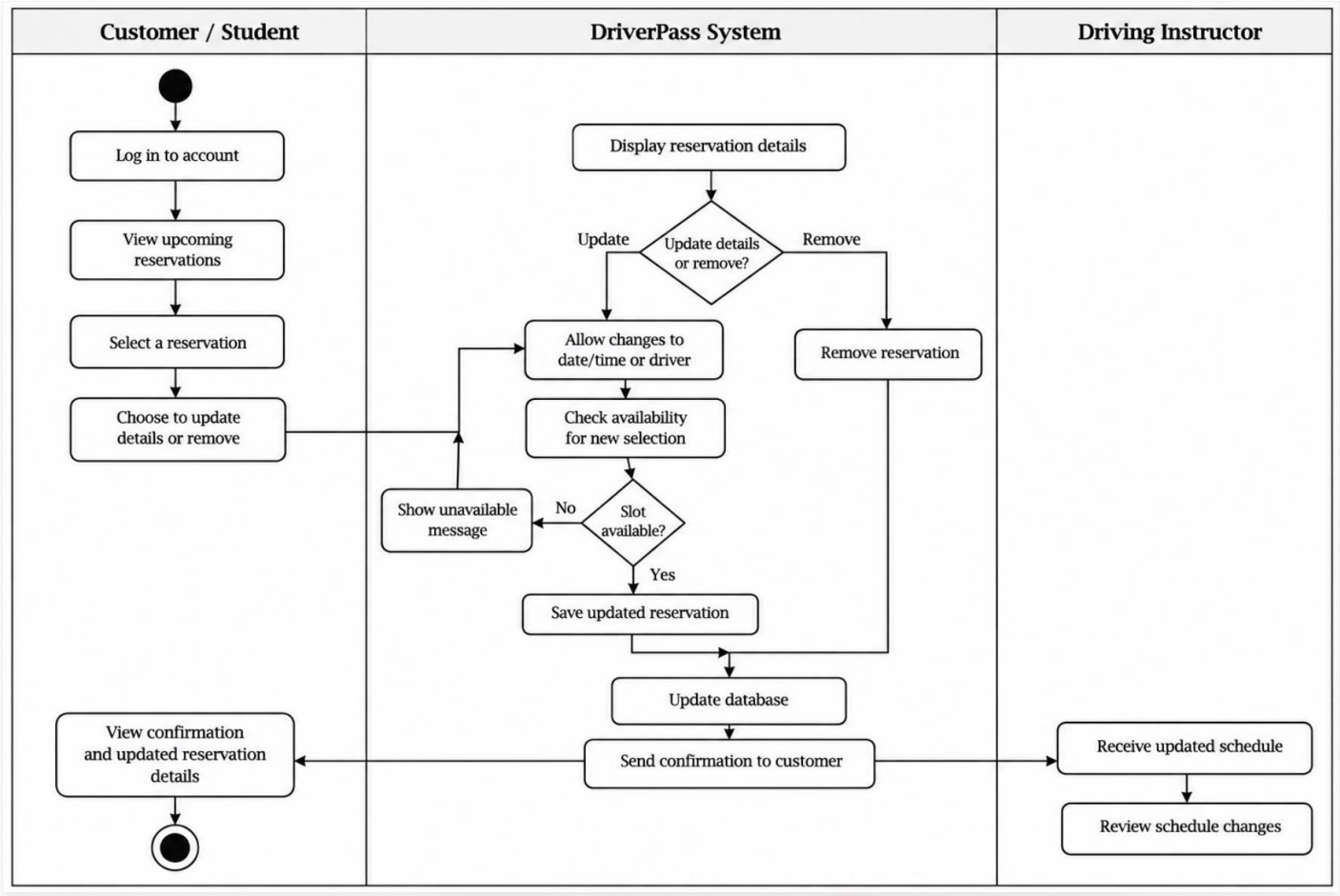


UML Activity Diagrams

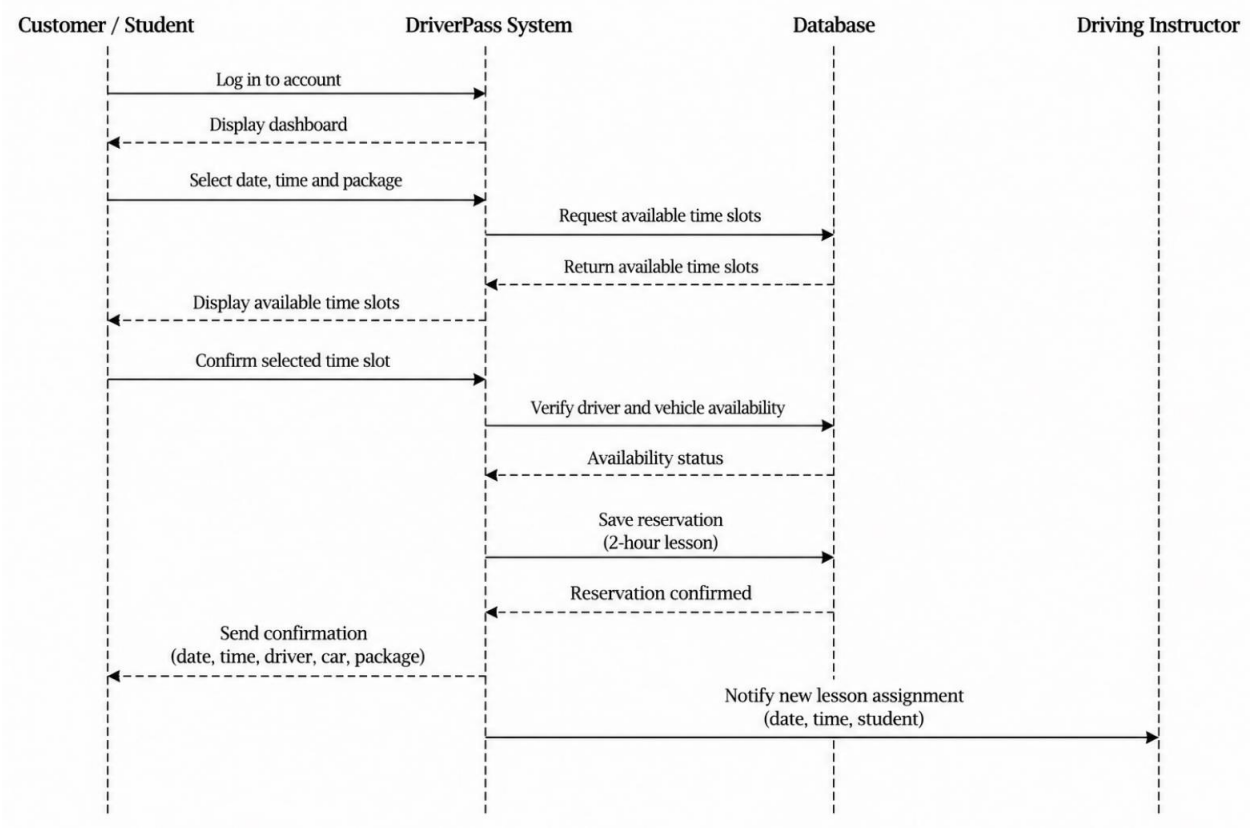
1. Schedule Driving Lesson


Customer/Student
DriverpassSystem
Driving Instructor
Log in to ACCount
Validate user Credentials
Show error message (retry login)
no
Credentials Valid?
Select preferred date, time and package(2-hour lesson)
Display available lesson slots
Check Availability of driver and vehicle for selected slot
Choose a different date and time
no
slot available?
View confirmation (lesson, details, date,time, driver,car, package)
Reserve driver, vehicle and time slot (2hour lesson)
Receive new lession assignment with schdule details
Create reservation and save to database
Review assigned lesson
Send confirmation to customer (with lesson details)

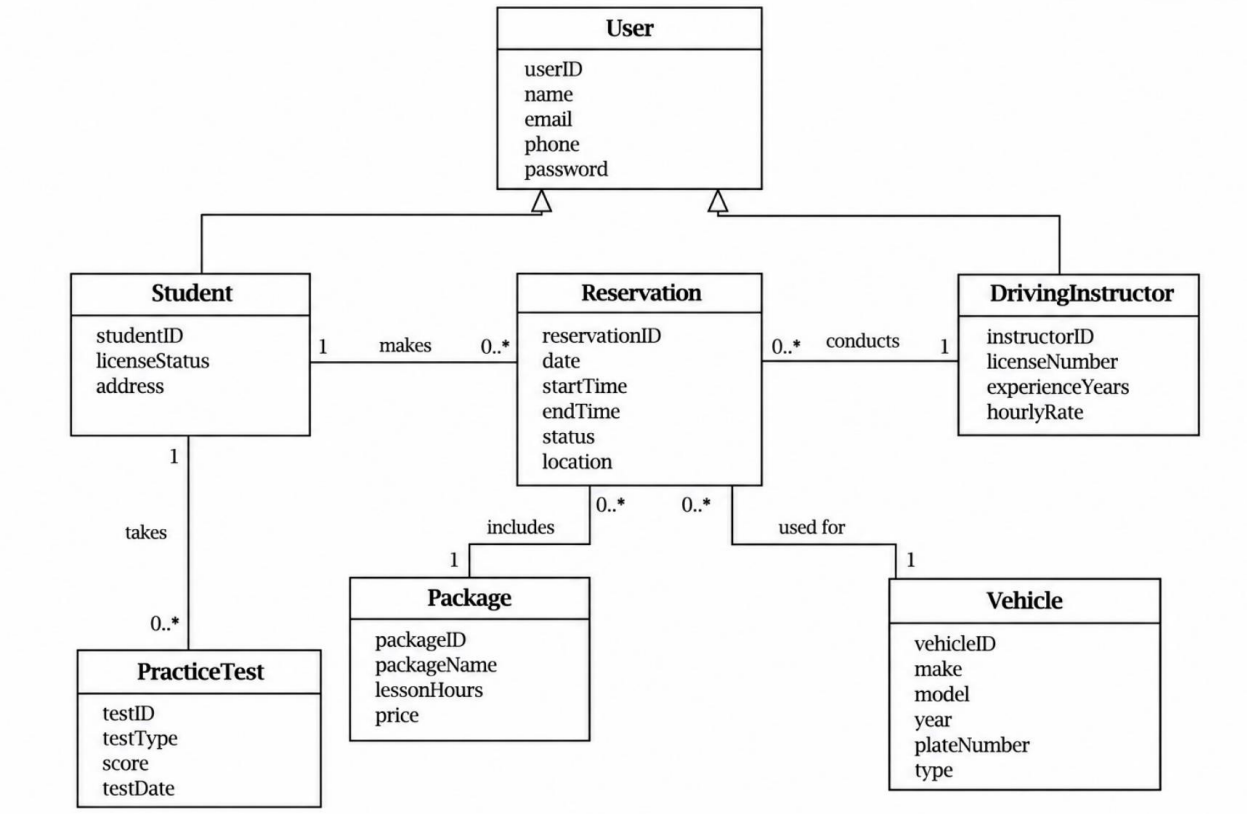
Cancel or Modify Reservation



UML Sequence Diagram



UML Class Diagram



Technical Requirements

Hardware Requirements

- The DriverPass system shall be accessible through desktop computers, laptops, tablets, and mobile devices with a stable internet connection.
- The system shall use cloud-based servers to host the web application, store customer and reservation data, and support regular data backups.
- DriverPass office staff shall have access to computers and reliable internet connectivity to manage customer information, driving lesson reservations, reports, and user accounts.

Software Requirements

- The system shall use a web-based application that can be accessed through a modern web browser.
- The system shall use a database to store customer information, user accounts, training packages, driving lesson reservations, vehicle details, instructor assignments, online class information, and practice test results.
- The application shall support customer registration, secure login, password resets, and role-based access for customers, the secretary, driving instructors, the owner, and the IT officer.
- The system shall allow authorized users to create, modify, and cancel driving lesson reservations while maintaining accurate records of the assigned customer, instructor, vehicle, date, and time.

- The system shall support online classes, practice tests, training package management, driver notes, activity reports, and updates to DMV-related training information.

Tools and Technologies

- UML diagrams created in Lucidchart shall be used to represent the system's structure, processes, interactions, and classes.
- Software development and database management tools shall be used to build, test, maintain, and update the application.
- The system shall provide reports that authorized users can view and export in an Excel-compatible format.
- The application shall support account management functions that allow the IT officer to reset passwords and block user access when necessary.

Infrastructure and Security

- The system shall use cloud-based infrastructure to make business information available to authorized users from different locations.
- Secure, encrypted connections shall protect information exchanged between users and the web application.
- Role-based access controls shall restrict system functions and information according to each user's responsibilities.
- Regular data backups and activity logs shall help protect business information and support recovery and accountability.
- The system shall record who creates, modifies, or cancels a reservation so the owner can review reservation activity reports.
- The system shall require an active internet connection to access current information and save changes to reservations and other records.
- The system shall support receiving updates to DMV rules, policies, and practice questions and notifying DriverPass when new information becomes available.